

Assignment 6: Data Scraping

Tracy Che

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

#1

```
library(tidyverse)
library(rvest)
library(lubridate)
library(here)
library(cowplot)
library(RColorBrewer)

getwd()
```

```
## [1] "/home/guest/tracyche"
```

```
here::i_am(
  "Assignments/TracyChe_A06_DataScraping.Rmd"
)

here()
```

```
## [1] "/home/guest/tracyche"
```

```
mytheme <- theme_gray() +
  theme(
    axis.text = element_text(color = "black"),
    legend.position = "top"
  )

theme_set(mytheme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2

the_url <- paste0(
  "https://www.ncwater.org/wudc/app/lwsp/report.php?",
  "pwsid=03-32-010&year=2024"
)

the_website <- read_html(the_url)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3

#name_css <- "div+ table tr:nth-child(1) td:nth-child(2)"
#pwsid_css <- "td tr:nth-child(1) td:nth-child(5)"
```

```
#ownership_css <- "div+ table tr:nth-child(2) td:nth-child(4)"
#maxday_css <- c("40.2400", "37.3800", "37.9100", "33.8100", "36.8400", "36.9000",
                #"36.5300", "38.0400", "33.8000", "37.9500", "35.2000", "32.2000")
```

```
the_name <- the_website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text(trim = TRUE)
```

```
the_pwsid <- the_website %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text(trim = TRUE)
```

```
the_ownership <- the_website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text(trim = TRUE)
```

```
the_maxday <- the_website %>%
  html_nodes('th~ td+ td') %>%
  html_text(trim = TRUE)
```

```
the_name
```

```
## [1] "Durham"
```

```
the_pwsid
```

```
## [1] "03-32-010"
```

```
the_ownership
```

```
## [1] "Municipality"
```

```
the_maxday
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

#4

```
Durham2024 <- data.frame(
  System = rep(the_name, 12),
  PWSID = rep(the_pwsid, 12),
  Ownership = rep(the_ownership, 12),
  MaxDay = as.numeric(the_maxday),
  Month = c(
    "Jan", "May", "Sep",
    "Feb", "Jun", "Oct",
    "Mar", "Jul", "Nov",
    "Apr", "Aug", "Dec"
  )
) %>%
mutate(
  Date = mdy(paste0(Month, "-1-2024"))
) %>%
arrange(Date)
```

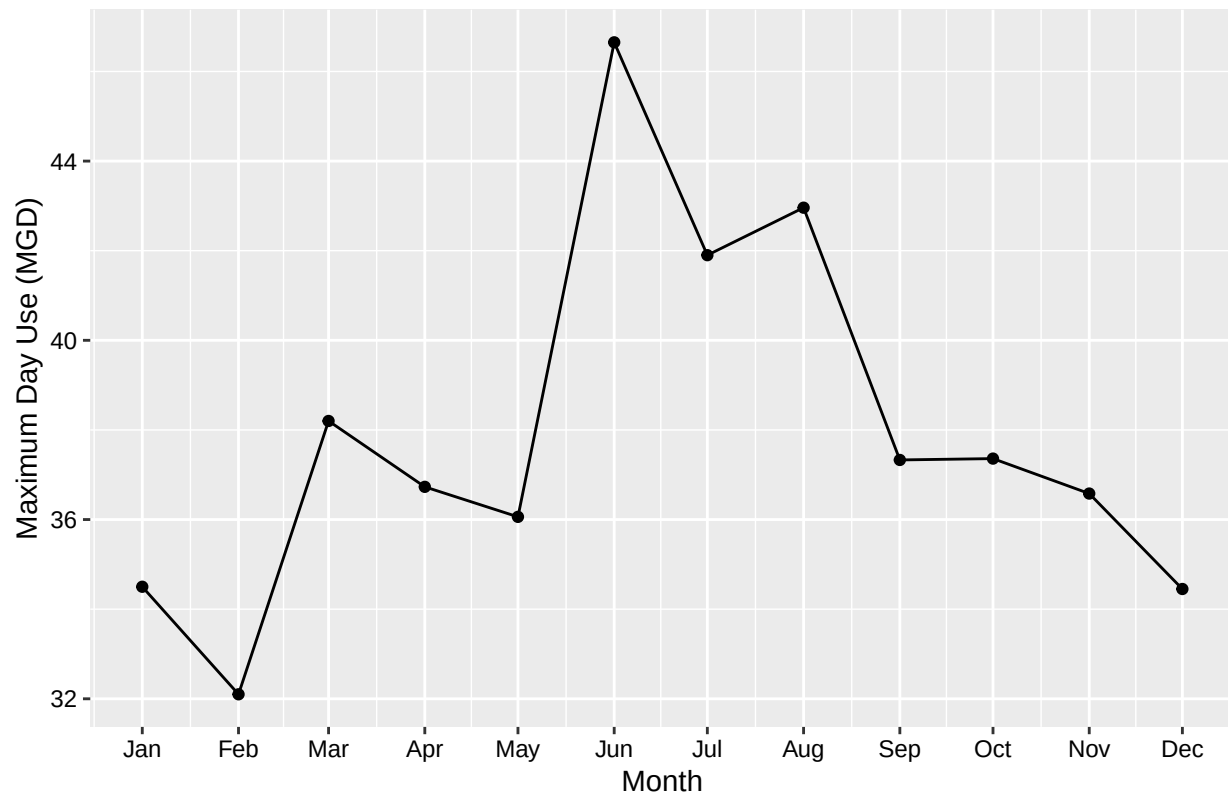
Durham2024

##	System	PWSID	Ownership	MaxDay	Month	Date
## 1	Durham	03-32-010	Municipality	34.50	Jan	2024-01-01
## 2	Durham	03-32-010	Municipality	32.10	Feb	2024-02-01
## 3	Durham	03-32-010	Municipality	38.20	Mar	2024-03-01
## 4	Durham	03-32-010	Municipality	36.73	Apr	2024-04-01
## 5	Durham	03-32-010	Municipality	36.06	May	2024-05-01
## 6	Durham	03-32-010	Municipality	46.65	Jun	2024-06-01
## 7	Durham	03-32-010	Municipality	41.90	Jul	2024-07-01
## 8	Durham	03-32-010	Municipality	42.96	Aug	2024-08-01
## 9	Durham	03-32-010	Municipality	37.33	Sep	2024-09-01
## 10	Durham	03-32-010	Municipality	37.36	Oct	2024-10-01
## 11	Durham	03-32-010	Municipality	36.58	Nov	2024-11-01
## 12	Durham	03-32-010	Municipality	34.45	Dec	2024-12-01

#5

```
Durham2024 %>%
  ggplot(aes(x = Date, y = MaxDay)) +
  geom_line() +
  geom_point() +
  scale_x_date(
    breaks = "1 month",
    date_labels = "%b"
  ) +
  labs(
    title = "Durham Maximum Daily Water Use in 2024",
    x = "Month",
    y = "Maximum Day Use (MGD)"
  )
```

Durham Maximum Daily Water Use in 2024



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

#6

```
scrape.it <- function(PWSID, year){

  # Create the URL
  the_url <- paste0(
    "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=",
    PWSID,
    "&year=",
    year
  )
  # Read the website
  the_website <- read_html(the_url)

  # Scrape the data
  the_name <- the_website %>%
```

```

html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
html_text(trim = TRUE)

the_ownership <- the_website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text(trim = TRUE)

the_maxday <- the_website %>%
  html_nodes("th~ td+ td") %>%
  html_text(trim = TRUE)

# Create dataframe
the_df <- data.frame(
  System = rep(the_name, 12),
  PWSID = rep(PWSID, 12),
  Ownership = rep(the_ownership, 12),
  MaxDay = as.numeric(the_maxday),
  Month = c(
    "Jan", "May", "Sep",
    "Feb", "Jun", "Oct",
    "Mar", "Jul", "Nov",
    "Apr", "Aug", "Dec"
  ),
  Year = rep(year, 12)
) %>%
mutate(
  Date = mdy(paste0(Month, "-1-", Year))
) %>%
arrange(Date)

# Return dataframe
return(the_df)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7

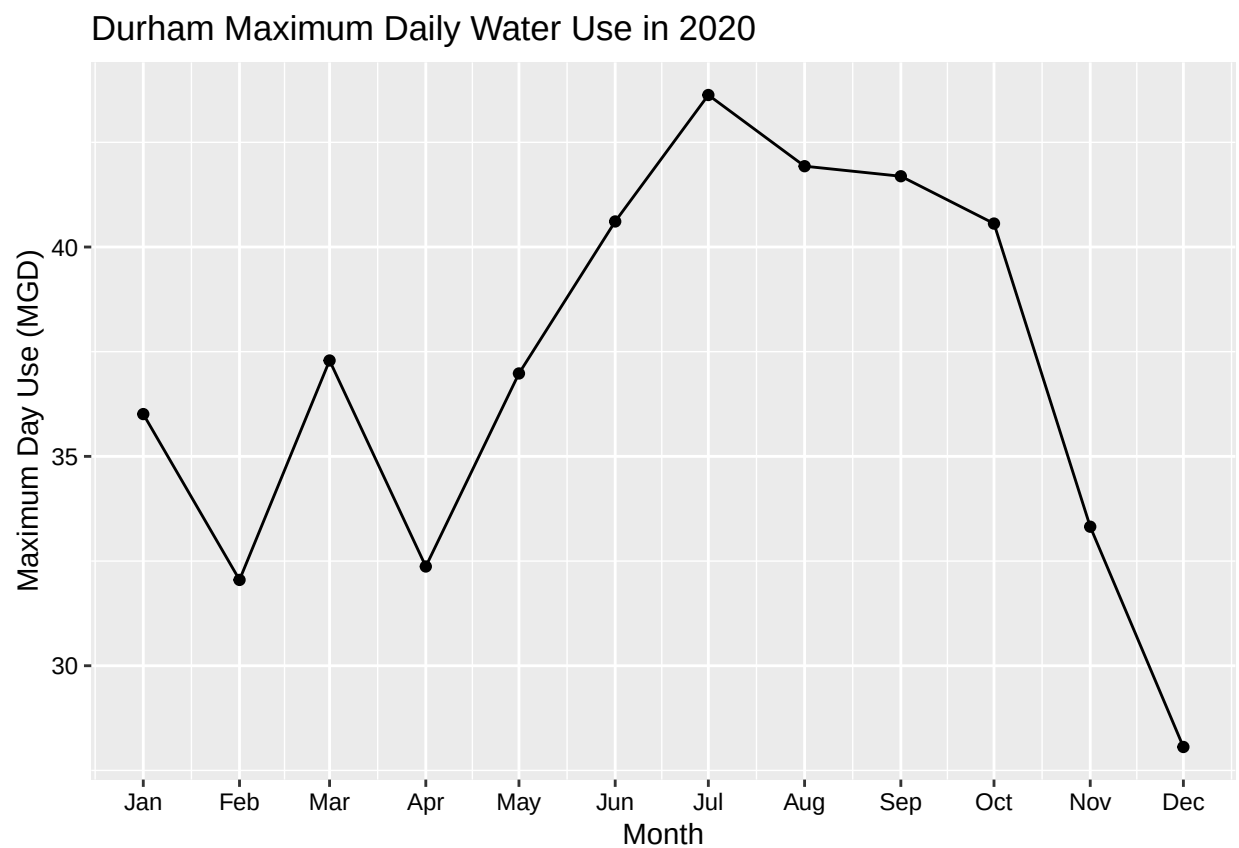
Durham2020 <- scrape.it("03-32-010", 2020)

glimpse(Durham2020)

## Rows: 12
## Columns: 7
## $ System    <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", ~
## $ PWSID     <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-0~
## $ Ownership <chr> "Municipality", "Municipality", "Municipality", "Municipalit~
## $ MaxDay    <dbl> 36.01, 32.05, 37.29, 32.37, 36.98, 40.61, 43.63, 41.93, 41.6~
## $ Month     <chr> "Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep~
## $ Year      <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, ~
## $ Date      <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 2020-05-01,~

```

```
Durham2020 %>%
  ggplot(aes(x = Date, y = MaxDay)) +
  geom_line() +
  geom_point() +
  scale_x_date(
    breaks = "1 month",
    date_labels = "%b"
  ) +
  labs(
    title = "Durham Maximum Daily Water Use in 2020",
    x = "Month",
    y = "Maximum Day Use (MGD)"
  )
)
```



8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

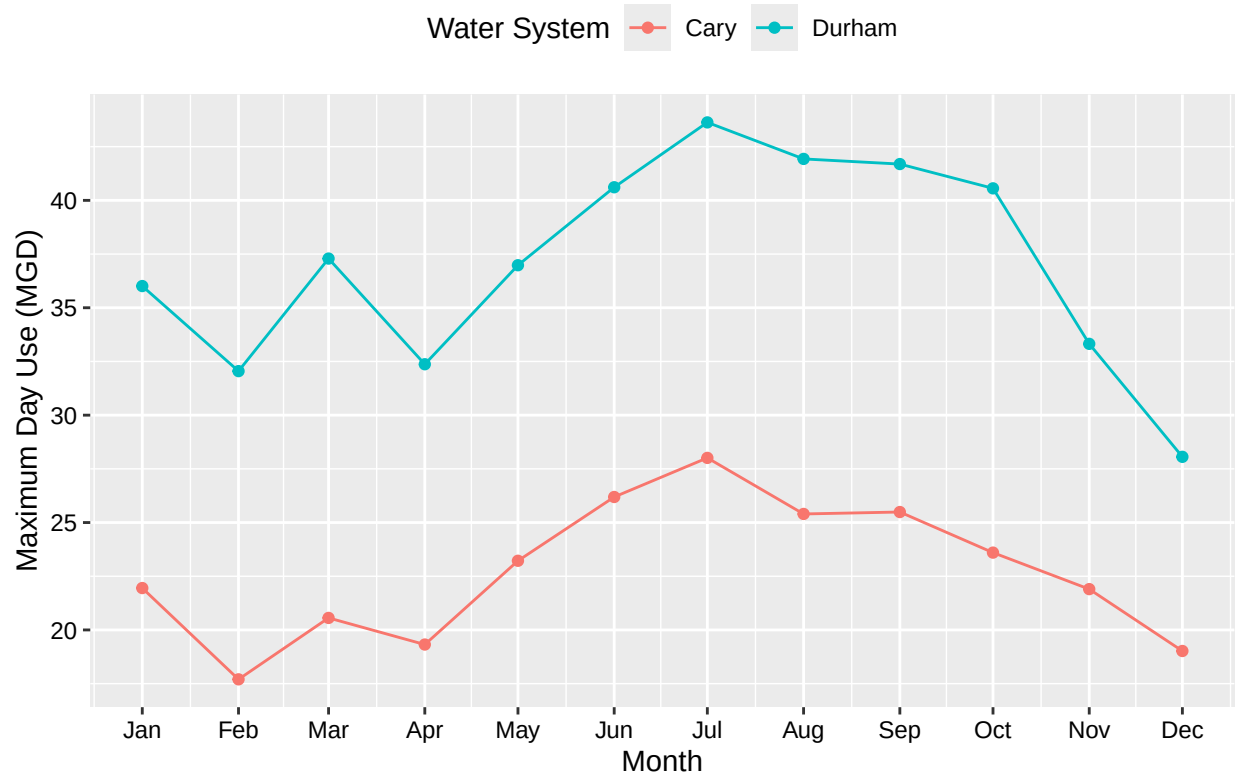
```
#8

Cary2020 <- scrape.it("03-92-020", 2020)

water2020 <- bind_rows(
  Durham2020,
  Cary2020
)
```

```
)
water2020 %>%
  ggplot(aes(x = Date, y = MaxDay, color = System)) +
  geom_line() +
  geom_point() +
  scale_x_date(
    breaks = "1 month",
    date_labels = "%b"
  ) +
  labs(
    title = "Durham and Cary Maximum Daily Water Use in 2020",
    x = "Month",
    y = "Maximum Day Use (MGD)",
    color = "Water System"
  )
)
```

Durham and Cary Maximum Daily Water Use in 2020



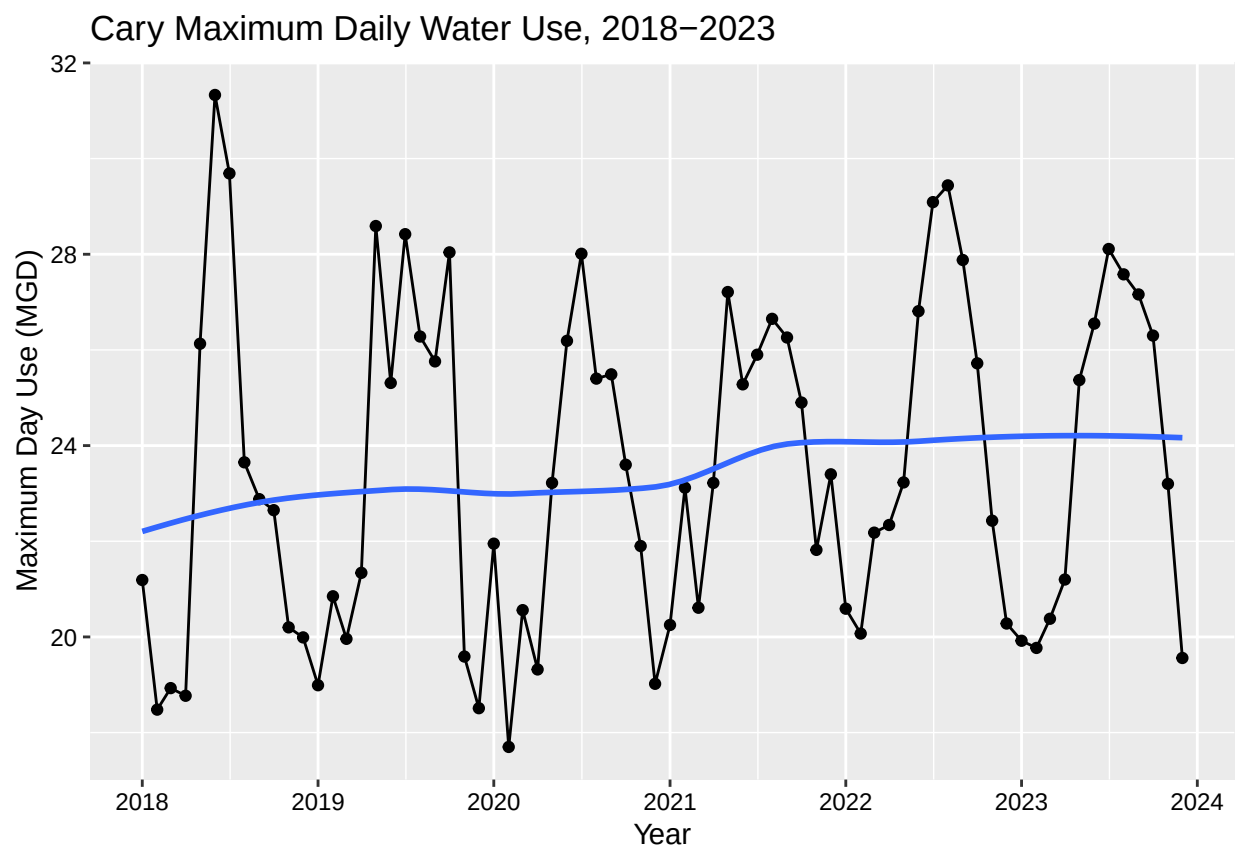
9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
Cary.multiyear <- map2(
  rep("03-92-020", 6),
  2018:2023,
  scrape.it
) %>%
  bind_rows()

Cary.multiyear %>%
  ggplot(aes(x = Date, y = MaxDay)) +
  geom_line() +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE) +
  scale_x_date(
    breaks = "1 year",
    date_labels = "%Y"
  ) +
  labs(
    title = "Cary Maximum Daily Water Use, 2018-2023",
    x = "Year",
    y = "Maximum Day Use (MGD)"
  )
)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes. Cary has a slight increasing trend of water usage over the years. Although there are seasonal variations each year, the smooth line shows slight increase from around 22 MGD to over 24 MGD.
>